



Secure and efficient mutual authentication framework for IoT: a blockchain-integrated graph neural network approach

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Abstract

The growth in the Internet of Things (IoT) demands the development of a secure and efficient authentication framework for safeguarding Machine-to-Machine (M2M) communication. Traditional centralized authentication systems face vulnerabilities, including single points of failure and data tampering, while the heterogeneity of IoT devices creates challenges in energy consumption and complicates anomaly detection. This research proposes MABF-IoT, a Blockchain-Integrated Graph Neural Network (GNN) framework that addresses authentication challenges in M2M communication within IoT environments. MABF-IoT leverages digital device fingerprinting for mutual authentication and employs a GNN-based anomaly detection model with a weighted aggregation function that adapts to evolving threats. The framework incorporates a Proof-of-Stake Consensus mechanism to reduce computational overhead. The comprehensive evaluation demonstrates that MABF-IoT outperforms existing frameworks (SAF, BAS, PuF, BPPS) in terms of recall, accuracy, and precision, with values ranging from 92% to 95%. Quality of Service (QoS) metrics indicate reduced energy consumption, authentication, and identification delay, while throughput and packet delivery ratio increase by 8.5% and 4.9%, respectively, compared to other models. The framework offers a scalable, efficient, and secure authentication solution for resource-constrained IoT environments, thereby enhancing the security of M2M communication.

Keywords Anomaly detection · Blockchain · Graph neural network · IoT security · Mutual authentication · Quality of services

1 Introduction

The rapid growth of the Internet of Things (IoT) has revolutionized digital ecosystems by enabling Machine-to-Machine (M2M) communication across diverse applications, including smart cities, industrial automation, healthcare, and intelligent transportation systems. As more IoT devices connect, security vulnerabilities increase, especially in areas such as device authentication, anomaly detection, and trust management. IoT devices are often placed in untrusted environments, which demands a secure and tamper-proof identity verification process. Traditional security models lack tamper-proof authentication logs, making them vulnerable to man-in-the-middle (MITM) and replay attacks [1]. The lack of transparency in identity validation and the challenge of scalability in the dynamic IoT landscape—where devices frequently join and leave networks—complicate the detection of unauthorized device access and credential forgeries [2, 3]. Conventional IoT authentication methods depend on a trusted central authority to verify

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device identities [4]. This setup is susceptible to single points of failure, cybersecurity threats, and data breaches, leading to network-wide vulnerabilities if the central server is compromised [5–7]. The widespread deployment of connected devices and reliance on third parties cause authentication delays, identity spoofing, and high cryptographic computation overhead, which can limit their use in battery-powered IoT devices [8–10].

Current anomaly detection systems for M2M communications rely on predefined attack signatures and fixed feature extraction methods, which are unable to adapt to emerging security threats. As a result, they struggle to identify new attack patterns, such as Zero-day exploits and adversarial manipulations [11–14]. The difficulty in detecting unauthorized connected devices is exemplified by the ongoing attack on the NASA Jet Propulsion Laboratory, where an unauthorized device accessed 23 critical mission-related files [15]. This incident highlights the importance of identifying and authenticating connected devices to prevent unauthorized access to the network. High-dimensional IoT traffic data requires adaptive models to enhance threat classification.

Blockchain-based authentication has become a decentralized alternative, utilizing distributed ledger technology (DLT) to create tamper-proof records of authentication. However, existing blockchain implementations encounter scalability problems due to high computational and storage demands, especially in resource-limited IoT environments [16]. Recent studies have also introduced deep learning-based threat detection models, but these methods are often computationally intensive, making them unsuitable for low-power IoT devices [17].

The main drawbacks of the authentication paradigm include its reliance on a trusted authority, the use of cryptographic keys for authentication and device identification, rigid input representations, lack of transparency, multi-step pre-processing, a single point of failure, and the absence of mutual authentication. There is a need for a scalable, intelligent, and energy-efficient authentication paradigm that supports decentralized security and learning-based threat detection protocols. MABF-IoT (Mutual Authentication Blockchain Framework for IoT), a blockchain-integrated graph neural network (GNN)-based approach that enhances mutual authentication, anomaly detection, and computational efficiency, is proposed by this study.

The proposed MABF-IoT framework introduces a decentralized authentication system for IoT devices using blockchain and GNN-based anomaly detection. MABF-IoT creates a device fingerprint from live device metadata, which is securely stored and authenticated via a blockchain ledger. This ensures tamper-proof and verifiable device authentication without centralized identity management. When an attack or unauthorized request is detected, the

system employs weighted aggregate anomaly detection methods with GNN to identify and respond to threats. It also uses Proof of Stake as its consensus mechanism. The energy-efficient framework manages multiple transactions without requiring additional computational resources. The innovation of this research lies in utilizing real-time device metadata to generate a device fingerprint for authentication. The unique key produced through this process cannot be replicated. Additionally, the study compares the proposed MABF-IoT model with other similar models, as shown in Table 1 (Sect. 2). The table demonstrates how MABF-IoT addresses the limitations of existing approaches and how they can be improved. Another innovation is the application of GNN for anomaly detection. The weighted aggregation function of the GNN allows the framework to adapt to evolving threat environments without extensive retraining. Details of this process are explained in Sect. 3.

The primary contribution of the study is:

1. MABF-IoT is a new framework that combines blockchain technology and GNN to secure IoT systems.
2. Implementation of a decentralized authentication system that generates unique device fingerprints from actual device data.
3. Using GNN-based anomaly detection with weighted aggregation functions.
4. Use the PoS consensus mechanism to create an energy-efficient platform capable of handling multiple transactions without needing extra computational power.

This research presents a comprehensive solution to the limitations of existing IoT authentication systems, including the elimination of single points of failure, reduction of computational overhead on resource-constrained devices, and increased transparency in identity verification.

The paper is structured as follows: Sect. 2 reviews related work in this field. Section 3 introduces the proposed model, including its system, security features, and mutual authentication mechanism. Section 4 details the experimental setup, including datasets, environmental configuration, performance metrics, and the tools and technologies used for evaluation. Section 5 presents the results and discusses the experimental analysis. Finally, Sect. 6 summarizes the main findings and their implications, concluding the overall work of the paper.

2 Related work

Several blockchain-based solutions have been proposed to address the authentication challenges in IoT networks. However, these frameworks exhibit critical limitations in

Table 1 Summarize the comparison of various blockchain-based schemas for IoT

Framework	Methodology	Anomaly detection	Scalability	Cryptographic overhead	Computational efficiency	Adaptive security	Limitations
Singh & Park [18]	Blockchain-based trust management for smart enterprises	No	Limited	High	Moderate	No	Computationally expensive for IoT
Garba et al. [19]	Blockchain-based lightweight certificate authentication	No	Moderate	Low	High	No	Relies on static cryptographic keys
Zhou et al. [20]	Blockchain for secure image sharing	No	Limited	Moderate	Low	No	Not designed for IoT authentication
Yang et al. [21]	Blockchain-based identity verification for the Metaverse	No	Moderate	High	Low	No	No real-time intrusion detection
Wu et al. [22]	Blockchain-based trusted ADS-B service	No	Limited	High	Low	No	Not optimized for low-power IoT
Wu et al. [23]	Blockchain-based GNSS authentication	No	High	Moderate	Moderate	No	No anomaly detection mechanism
Zhang et al. [26]	Blockchain-based privacy-preserving authentication (DBCPA)	No	High	Moderate	High	No	No adaptive learning for evolving threats
Subramani et al. [28]	Blockchain-based anonymous authentication for Fog VANETs	No	Moderate	Low	High	No	No real-time anomaly detection
Karim et al. [30]	Blockchain-based secure data collection for IoV	No	High	Moderate	High	No	Lacks AI-based intrusion detection
Lin et al. [32]	Blockchain-based privacy-preserving authentication for VANETs	No	High	Low	High	No	No anomaly detection integration
El-Zawawy et al. [33]	Blockchain-based drone authentication for IoV	No	Moderate	Low	High	No	Limited scalability in IoT networks
Son et al. [34]	Blockchain-based lightweight UAV authentication	No	High	Low	High	No	No adaptive security model
Zhang et al. [35]	Blockchain-based access authentication for IIoT	No	High	Low	High	No	Lacks adaptive anomaly detection
Liu et al. [36]	Blockchain-based cross-domain authentication for IoT	No	High	Low	High	No	No AI-driven threat detection
Dong et al. [37]	Blockchain-based certificate-free cross-domain authentication	No	High	Low	High	No	No adaptive anomaly detection
Park et al. [38]	Blockchain-based privacy-preserving authentication for Smart Grid	No	High	Low	High	No	Lacks real-time authentication updates
Deebak et al. [39]	Lightweight blockchain-based mutual authentication for AI-IoT	No	Moderate	Low	High	No	Not optimized for large-scale IoT networks
Proposed MABF-IoT	Blockchain-Integrated Graph Neural Network (GNN) framework	Yes (GNN-Based Anomaly Detection)	High	Low	Optimized	Yes (Adaptive to Dynamic Threats)	Requires real-world deployment for further validation

terms of scalability, computational efficiency, and security guarantees. This section provides a comparative analysis of the most notable approaches and their limitations.

Singh and Park [18] introduce TaLWaR, a blockchain-based trust management scheme that leverages blockchain's transparency and immutability to foster trust within augmented intelligence environments in enterprises. Garba et al. [19] focus on lightweight certificate authentication in IoT applications, and Zhou et al. [20] present a blockchain-based solution for secure and efficient secret image sharing. Yang et al. [21] propose a secure authentication framework to ensure avatar traceability, thereby expanding the scope

of blockchain applications beyond traditional IoT environments. Wu et al. [22] introduce ADS-Bchain, a blockchain-based service scheme for aviation surveillance where data integrity and security are paramount. Additionally, Wu et al. [23] explore the utilization of blockchain to authenticate navigation messages, thereby enhancing the integrity and security of critical communication systems. Tao et al. [24] present B-DSPA, a dynamically scalable privacy-preserving authentication scheme. Xie et al. [25] propose a blockchain-based protocol for V2I handover authentication and V2V broadcast authentication, while Zhang et al. [26] introduce a dual blockchain-assisted authentication framework. Zhang et

al. [27] propose a blockchain-based authentication scheme, highlighting the versatility of blockchain applications across different transportation modes. Subramani et al. [28] focus on fog-based VANETs, proposing a blockchain-based physically secure and privacy-aware anonymous authentication scheme. Liu et al. [29] introduce CPAHP, a conditional privacy-preserving authentication scheme with a hierarchical pseudonym for 5G-enabled Internet of Vehicles (IoV) systems, demonstrating the convergence of blockchain with 5G technology in vehicular networks. Karim et al. [30] present BSDCE-IoV, a blockchain-based scheme for secure data collection and exchange in IoV within a 5G environment. Dwivedi et al. [31] propose B-HAS, a blockchain-assisted handover authentication protocol, while Lin et al. [32] introduce EBCPA, an efficient blockchain-based authentication system. El-Zawawy et al. [33] explore the authentication of drone-assisted IoV using elliptic curve cryptography and blockchain. Son et al. [34] combine blockchain with physical unclonable functions (PUF) to design a secure and lightweight authentication scheme for UAV-enabled intelligent transportation systems. Zhang et al. [35] propose RRV-BC, a random reputation voting mechanism combined with blockchain, to enhance access authentication in IIoT. Liu et al. [36] delve into cross-domain authentication management for IoT devices, demonstrating the effectiveness of blockchain in managing complex IoT ecosystems. Dong et al. [37] contribute to industrial internet security by developing a blockchain-based, certificate-free, cross-domain authentication mechanism. Park et al. [38] focus on smart grid environments, presenting a blockchain-enabled, privacy-preserving scheme for demand-response management that addresses security and privacy challenges in smart grids. Deebak et al. [39] propose a lightweight blockchain-based remote mutual authentication system and sustainable computing systems, enhancing security and sustainability within IoT environments.

Table 1 summarizes and compares the key aspects of the reviewed framework, focusing on methodology, computational efficiency, scalability, and their limitations. One of the primary shortcomings of most existing frameworks is the lack of anomaly detection and adaptive security measures. Solutions such as BAS, SAF, PuF, and BPPS lack real-time anomaly detection, have high computational costs, and incur cryptography overhead.

The proposed MABF-IoT framework employs a fully decentralized, automated approach through the use of smart contracts. It incorporates a GNN-based anomaly detection model that safeguards data, maintains security, and grants complete control over identities across the IoT ecosystem, decreasing dependence on third-party authorities and improving privacy. Using device digital fingerprinting for

security and authenticity addresses the mutual authentication limitations of existing authentication frameworks.

3 Proposed methodology

The MABF-IoT framework improves security, scalability, and efficiency in IoT authentication and anomaly detection. Here are its main features:

1. Decentralized authentication using Blockchain.
 - Eliminates Single Points of Failure: MABF-IoT leverages blockchain-based distributed identity management, preventing server-dependent security failures.
 - Tamper-Proof and Transparent Identity Verification: Every authentication record and digital fingerprint is kept on an immutable, secure blockchain ledger to maintain data integrity and non-repudiation.
 - Smart Contract-Based Authentication: Employing blockchain smart contracts for trustless authentication and minimizing dependency on third-party authorities.
 - Blockchain Consensus Optimisation: Employing an improved *PoS* mechanism to alleviate the heavy computation.
2. GNN-Based Anomaly Detection.
 - Real-Time Threat Detection: Incorporates GNN to identify advanced cyber threats.
 - Adaptive learning capabilities: Apply weighted graph-based feature aggregation to enhance anomaly detection effectiveness repeatedly.
 - Fewer False Positives: GNN-based contextual analysis improves the precision of attack classification by lowering false alerts.
3. Digital Fingerprinting for Secure Identity Management.
 - Unique Device Identification: Produces dynamic digital fingerprints for IoT devices from current-state metadata.
 - Resistant to Attacks: Unlike cryptographic keys, device fingerprints are resistant to credential spoofing and unauthorized device cloning.
4. Secure Mutual Authentication for Machine-to-Machine (M2M) Communications.
 - Two-Way Authenticating Process: This process ensures that both IoT devices and the networking infrastructure authenticate to one another before communicating.

Table 2 Symbol table

S. No.	Notation	Meaning	S. No.	Notation	Meaning
1.	av	Aggregated feature vector	16.	DI	Device information
2.	v	IoT devices as a node	17.	DF	Device fingerprint
3.	e	Connection between nodes (Edges)	18.	BA	Blockchain address
4.	$N(v)$	Set of neighboring nodes	19.	N	Nonce values
5.	Xv'	Updated feature vector	20.	Req	Request
6.	w	Weight	21.	$Conn$	Connection
7.	b	Biases	22.	$Meta$	Metadata
8.	Lv	Training model	23.	$Exist$	Existence
9.	P	Probability	24.	$Verify$	Verification
10.	$Y(v, y)$	Node true label	25.	$Mapp$	Mapping
11.	y, j	Classes for different scenarios	26.	$Valid$	Validation
12.	DF_x	Digital fingerprint for device x	27.	Gen	Generation
13.	ID	Identifier	28.	T	True
14.	PU	Public Key	29.	F	False
15.	BS	Blockchain Server	30.	eq	Equation

This section outlines the system, authentication design, and security framework for the MABF-IoT framework to enable

the application of its trust mechanism for registration and device identification throughout the network. Table 2 provides the notations used and their roles for the framework.

The MABF-IoT methodology integrates blockchain-based authentication, GNN-powered anomaly detection, and device digital fingerprinting to establish a secure, scalable, and efficient IoT mutual authentication mechanism.

3.1 Blockchain-enabled system architecture

The proposed system consists of four phases: device connection, anomaly detection, storage, and authentication. Figure 1 illustrates the architecture of the model. During the device connection phase, IoT devices connect to the network through gateways. Each gateway oversees a cluster of IoT devices that generate data according to specific requirements.

The authentication phase uses gateways with multiple features, including low transmit power, fast and reliable transmission, and extensive computing power. Gateways perform heavy processing tasks that IoT devices, due to their limited capabilities, cannot handle, thereby reducing server load and minimizing network delay. In the MABF-IoT, the authentication process uses real-time device metadata to create digital fingerprinting, a unique digital ID for each IoT device. The storage phase offers tamper-proof,

Fig. 1 Model architecture for the MABF-IoT [16]



distributed, and efficient data storage for the IoT environment. The model uses the PoS mechanism to reach consensus among the nodes.

3.2 GNN-based adaptive anomaly detection

The GNN model's ability to represent traffic in a graph format and aggregate neighborhood information makes it an effective tool for advanced network analysis, with great potential for IoT device management and security. The architecture of the GNN model, shown in Fig. 2, is designed to process and analyze the graph representation of traffic systematically. The illustration offers a clear overview of the model's layers and the flow of information within the system. Each node corresponds to a feature vector containing the characteristics of the traffic at that particular node. By efficiently leveraging the graph's structural knowledge and interactions, the aggregation process enables the GNN to learn comprehensive patterns within the network and distinguish between regular and malicious activity.

GNN Architecture Components:

1. Input Layer:

This layer represents the initial input, including the feature vectors of the IoT devices.

2. Aggregation Layer:

Aggregates information from the neighboring nodes, modeled using Eq. 1:

$$av = AGGREGATE(\{Xu : u \in N(v)\}) \quad (1)$$

where, av is the aggregated feature vector of the node v , and $N(v)$ represents the set of neighboring nodes of v sets.

3. Update Layer:

Updates the feature vectors of the nodes based on the aggregated information using Eq. 2:

$$Xv' = Update\left(\sum_{i=1}^{N(v)} X(i, v) * w(a(i, v)) + b(i, v)\right) \quad (2)$$

where, Xv' is the updated feature vector of the node v sets, w & b represent the weights & biases.

4. Hidden Layers:

These layers iteratively apply these aggregation and update functions across all nodes in the graph for a specified number of layers or iterations in real-time scenarios. These layers perform additional transformations and learning, enabling the capture of complex patterns within the data.

5. Output Layer:

Produces the final feature representations for each node. This layer feeds into the classification and loss calculation components.

6. Cross-Entropy Loss Function:

Used to train the model by minimizing the difference between the predicted and actual class labels using Eq. 3:

$$Lv = - \sum_y Y(v, y) * \log(P(Yv = y | Xv')) \quad (3)$$

where, $Y(v, y)$ is the true label of the node v for class y sets.

7. Classification Layer:

Classifies nodes into different categories, such as normal or abnormal, using Eq. 4:

$$P(Yv = y | Xv') = \frac{e(Wy \cdot Xv')}{\sum_j e(Wj \cdot Xv')} \quad (4)$$

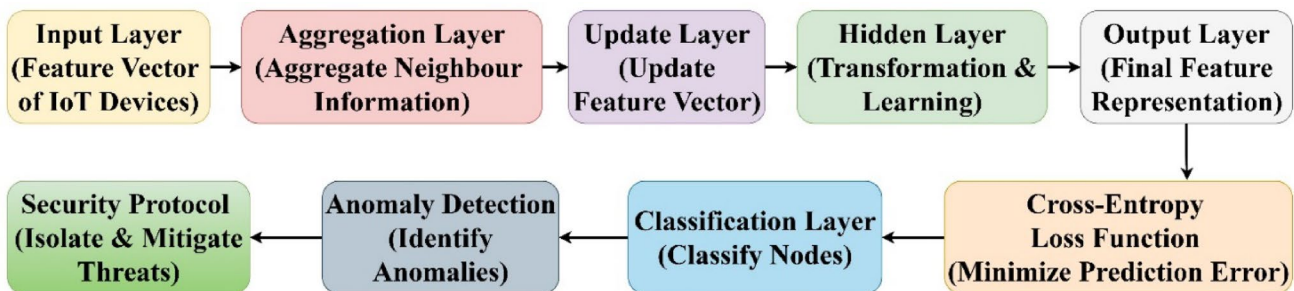


Fig. 2 GNN architecture of the MABF-IoT

where, $P(Yv = y | Xv')$ is the probability that node v belongs to the class y (e.g., normal, brute force attack, etc.), Wy and Wj are the weight vectors associated with classes y and j , respectively, for different scenarios.

8. Anomaly Detection:

Identifies nodes whose predicted class labels deviate from standard patterns.

9. Security Protocol:

If an anomaly is detected, the system initiates a security protocol to isolate and mitigate the threat.

3.3 Device digital fingerprinting for secure identity management

MABF-IoT introduces a dynamic digital fingerprinting mechanism to prevent device spoofing and identity theft. Figure 3 shows the proposed MABF-IoT block architecture.

When a device initiates a request to join the network, it generates transaction sets. This transaction includes the *Device ID* represented via Eq. 5.

$$Tjoin = \{(Device\ ID)\} \quad (5)$$

The transition checks for the pre-occurrence of the device. If the device exists, the request is denied; otherwise, the gateway fetches the device metadata as shown in Eq. 6.

$$Fetch = \{(DI), (IP), (PU)\} \quad (6)$$

The system first checks if the device's metadata, including information such as its unique identifier and current state, is already in the network. If it is, the request is rejected. If not, the system assigns the device a blockchain address. The metadata and blockchain address are combined to generate a unique device fingerprint (DF) using hash (H) shown in Eq. 7.

$$Generate(DF) = H\{(Device\ ID)(DI), (IP), (PU), (BA)\} \quad (7)$$

The consensus function $PoS(T)$ returns a Boolean value indicating the validity of the transactions. Once a transaction is validated, it is securely added to a new block in the chain using Eq. 8.

$$Bnew = Bnew \cup \{Tjoin\} \cup \{Fetch\} \cup \{Generate(DF)\} \quad (8)$$

Where, $\cup$ represents the union operation, signifying the addition of the transaction to the blocks. The new block is securely linked to the existing blockchain using a cryptographic linkage process. This link is created by including the previous block $B(i-1)$ hash (H) in the current block $B(i)$ sets. The linking is represented via Eq. 9,

$$Bi = \{H(Bi-1), Tjoin, Fetch, Generate(DF)..\} \quad (9)$$

Each block includes a nonce value N to enhance security further. This nonce value is a random number used in the PoS algorithmic process. The miners solve a cryptographic puzzle to find the correct nonce that produces a hash value that meets the network's criteria when hashed with the block's contents. Once the nonce is found, the block is considered valid and added to the blockchain. The updated blockchain is represented via Eq. 10,

$$Blockchain = Blockchain \cup \{Bi\} \quad (10)$$

3.4 Secure mutual authentication for M2M communication

Switching to decentralized identity storage, when a device joins the network, its identifier, metadata, device fingerprint, and assigned blockchain address are recorded on the blockchain along with their mappings. Figure 4 illustrates the complete process flow of the MABF-IoT. Table 3 outlines the scenarios for creating and saving device fingerprints on the blockchain. This decentralized method removes reliance

Fig. 3 MABF-IoT framework block architecture

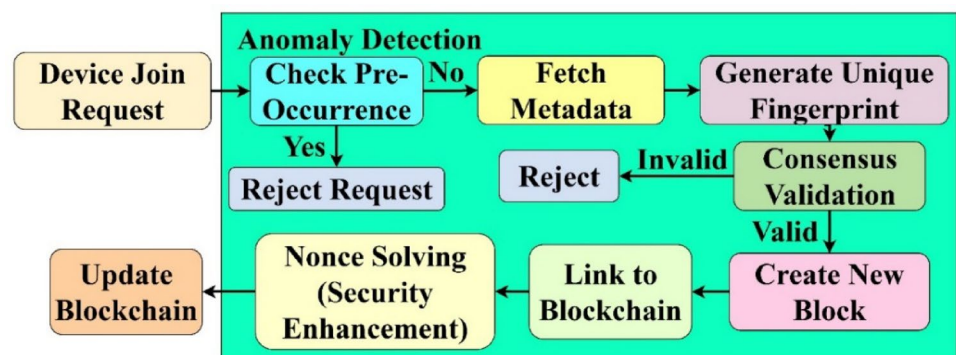
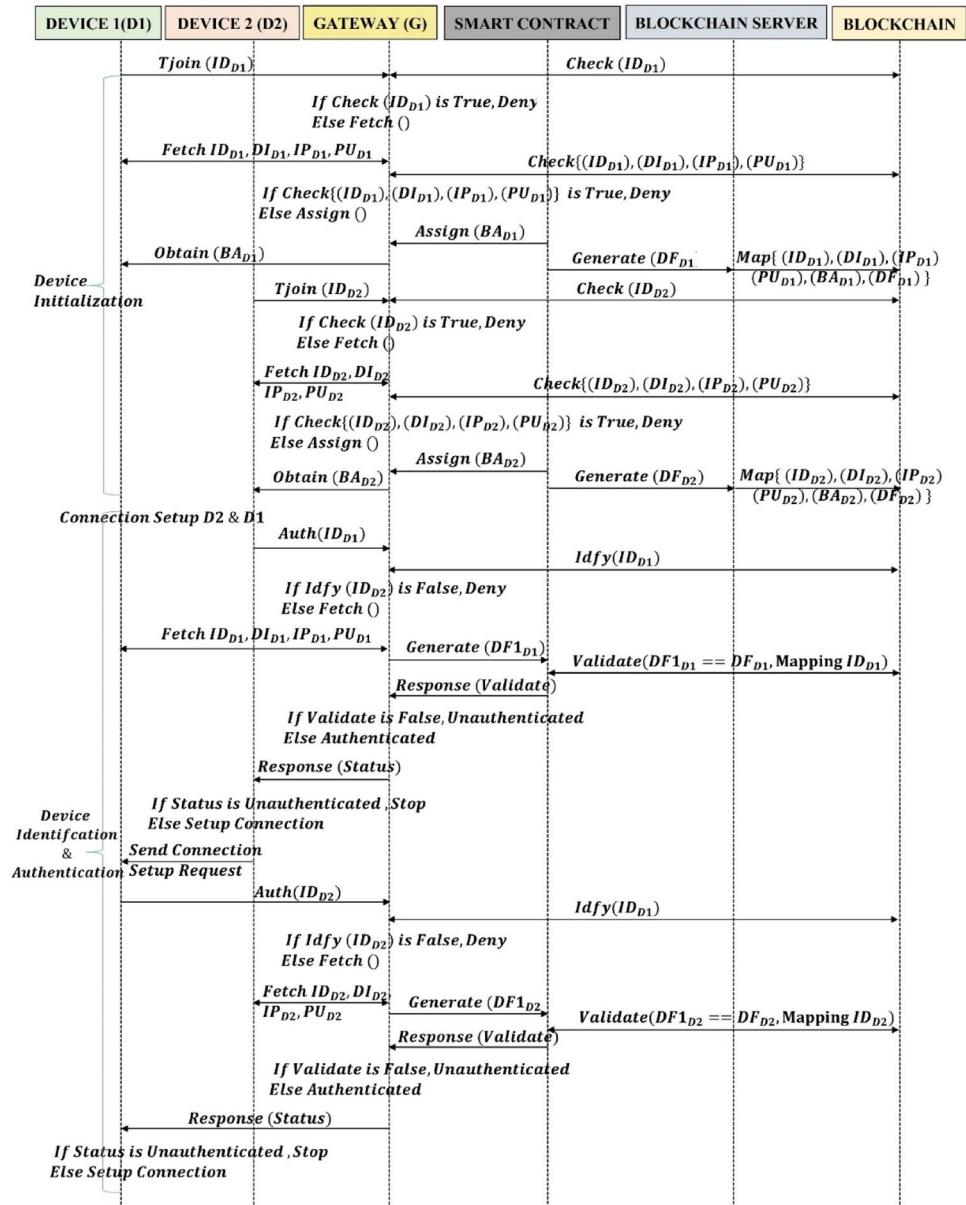


Fig. 4 Process flow diagram of the MABF-IoT [30]**Table 3** Decentralized Identity Storage

Device	Join Req.	Fetch Meta.	Pre-Exist. Verify	Assign BA	Mapp. Valid.	DF Gen.	Action
A	F	—	—	—	—	—	—
A	T	T	T	—	T	—	—
B	T	T	T	—	F	—	Block Device
B	T	T	F	T	—	T	Store on Blockchain

on centralized identity providers and enhances security levels.

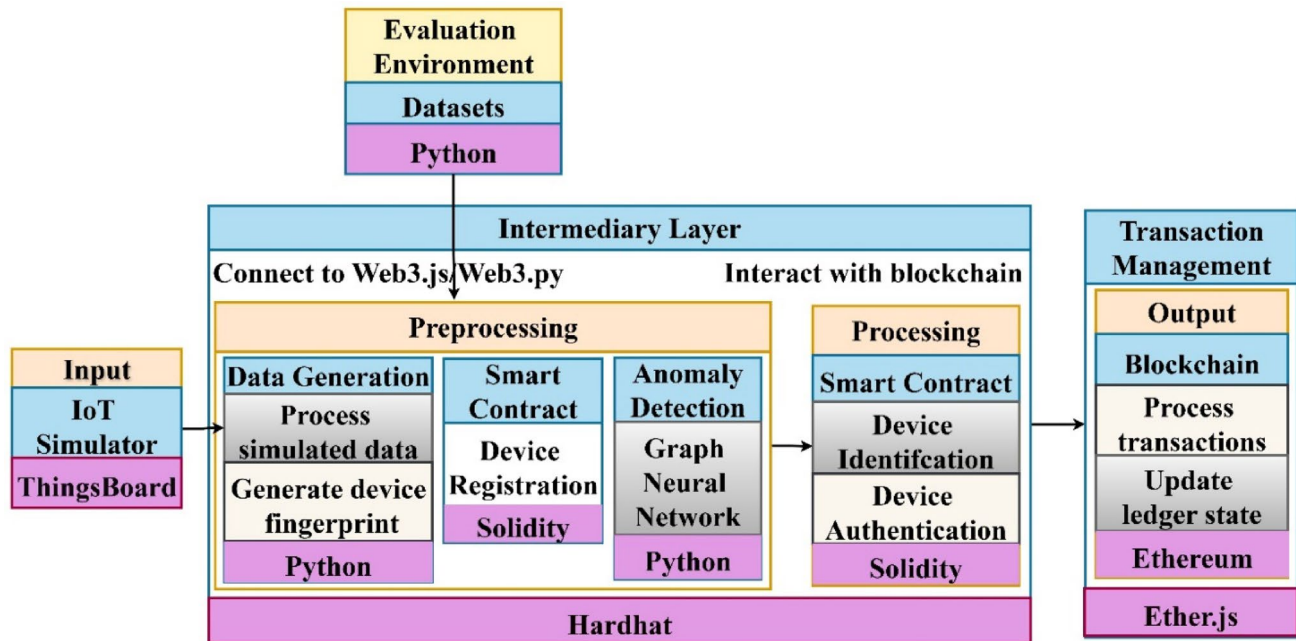
Considering a scenario where two IoT devices, *Device A* and *Device B*, are engaged in communications. *Device B* initiates the process by sending a message to *Device A* for different scenarios. As part of the authentication protocol, *Device B* generates an authenticated message for

Device A $Auth(ID_{D1})$ and sends this to the gateway. Gateway, upon receiving the message, identifies *Device A*'s $Idfy(ID_{D1})$ and validates *Device A*. This process is mirrored when *Device B* generates the connection setup request to *Device A*.

$Validate(DF_A)$ and $Validate(DF_B)$, represent the validation of the mapping of device identifiers, metadata,

Table 4 Digital Fingerprint Validation

Device	Conn. Req.	Fetch Meta.	Pre-Exist. verify	Mapp. Valid.	DF Valid.	Attack Scenario
A	F	–	–	–	–	–
A	T	T	T	T	T	No
B	T	T	T	F	–	Yes
B	T	T	T	T	F	Yes

**Fig. 5** Test Environment of MABF-IoT Framework

blockchain addresses, and device fingerprints of Devices A and B across different scenarios.

In the attack scenario, where the fingerprints may have been tampered with or forged, the verification fails, demonstrating the model's ability to detect and prevent potential security breaches as depicted in Table 4.

DF_A and DF_B represent the digital fingerprint generated for Devices A and B, respectively, for different use cases. In this exchange, the effectiveness of the mutual authentication is evident. Under normal conditions, the fingerprints are verified successfully, ensuring the authenticity of the messages.

Device mappings are securely stored and accurately retrieved in non-attack scenarios, enhancing the overall security architecture. However, under attack scenarios, correct mapping failure signifies the system's ability to identify and react to abnormal activities, thereby preventing unauthorized access.

This narrative will assist the readers through a user request scenario, detailing the outcomes at various stages of the authentication process, both under normal and attack conditions.

4 Experimental setup

A series of experiments was conducted under controlled conditions to assess the effectiveness of MABF-IoT in the IoT environment. Figure 5 shows the test environment for the experimental setup. The diagram illustrates the proposed framework for integrating IoT devices with a blockchain network. The IoT simulator mimics real-world devices, such as virtual cameras and biometric authenticators. Data generation introduces security measures, such as creating a device fingerprint during the preprocessing stage. Processing manages device identification and authentication, with rules defined in the smart contract. Finally, the processed transactions are recorded in the ledger. Ether.js enables seamless interaction with the Ethereum blockchain, and the intermediary layer connects the IoT devices to the blockchain. Further details of the experimental setups are provided below.

4.1 Datasets

The distinct datasets were employed to provide a comprehensive evaluation environment, shown in Table 5:

Table 5 Dataset

Datasets	Description	Key features	No. of records (Million)	Size (GB)
Login Data Set for Risk-Based Authentication [40, 41]	Synthesized real-world login behavior.	User ID, login timestamp, login status, etc.	33	1
BoT-IoT Dataset [42]	Contains realistic network traffic data with a mix of normal activities and various attack types, simulating an IoT network.	Timestamp, source IP, destination IP, source port, destination port, protocol, etc.	73	70
IoT Intrusion Detection [43]	The dataset is used to prepare the predictive model for detecting different kinds of intrusive attacks.	Weight, label, duration, protocol, rate etc.	1	47
IoT-23 Dataset [44]	Comprises network traffic captures from various IoT devices in benign and malicious scenarios.	Network flow data, including source/destination IP, ports, protocols, etc.	20	50

Table 6 Environment configuration

Component	Specification
IoT Nodes	Intel® Core™ i3-6006U processors, 512 MB – 1 GB RAM, WiFi connectivity, connected through the local network for communication.
Blockchain Network	Multiple nodes on separate virtual machines, Ubuntu 20.04 LTS, 8 GB RAM, simulating a decentralized environment.
Gateway	Intel® Core™ i5-8500 processors, 16 GB RAM, hosting the anomaly detection system, capable of real-time processing.
Blockchain Server	Intel® Xeon® processors, 32 GB of RAM, hosting the blockchain, ensuring high-performance and reliable transactions.

Table 7 Tools & technologies

Component	Specification	Purpose
Blockchain	Hardhat Framework	Streamlines Ethereum development tasks, enhancing efficiency and productivity.
Transaction Management	Ether.js	Facilitates seamless interaction with the Ethereum blockchain, including data querying interactions.
Programming	Python Solidity	For GNN implementation. Enables the definition of logic and behavior on the Ethereum blockchain, ensuring robust and secure execution.

4.2 Environmental configuration

The environment configuration hardware setup for MABF-IoT is depicted in Table 6:

4.3 Tools and technologies

The tools and technologies consisting of the software and languages used for MABF-IoT are depicted in Table 7:

4.4 Performance metrics

MABF-IoT's efficiency is evaluated and compared using primary and Quality of Service (QoS) performance metrics [45].

4.4.1 Primary performance metrics

The following are the primary performance metrics:

- **Precision:** Precision represents the proportion of true-positive identification.
- **Accuracy:** Accuracy represents the proportion of true results, including both true positive and true negative, in the total no. of identifications.
- **Recall:** The proportion of correctly identified positive results to the actual no. of positive results.
- **Specificity:** The proportion of correctly identified negatives to the actual no. of negatives.

4.4.2 QoS performance metrics

The following are the QoS performance metrics under the attack scenario:

- **Authentication Delay:** Authentication delay is the time required to authenticate the device on the network.
- **Identification Delay:** Identification delay is the time required to identify the device on the network.
- **Energy Consumption:** The Amount of energy consumed during the authentication process.
- **Throughput:** Throughput is the number of successful authentication requests processed per unit time, measured in kilobits per second (kbps).
- **Packet Delivery Ratio (PDR):** The PDR is the ratio of packets successfully delivered to the total number of packets sent.

The experimental setup enables a comprehensive evaluation of the proposed MABF-IoT framework in comparison to existing IoT authentication systems, highlighting its strengths and potential areas for improvement.

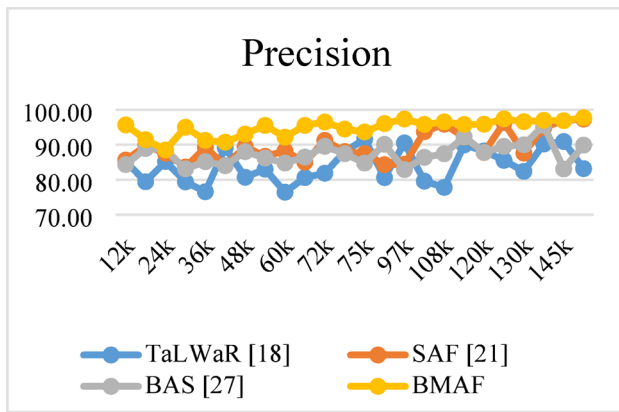


Fig. 6 Precision for authentication requests

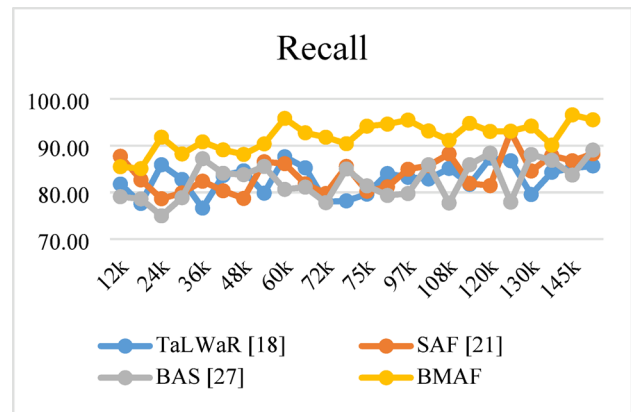


Fig. 8 Recall for authentication requests

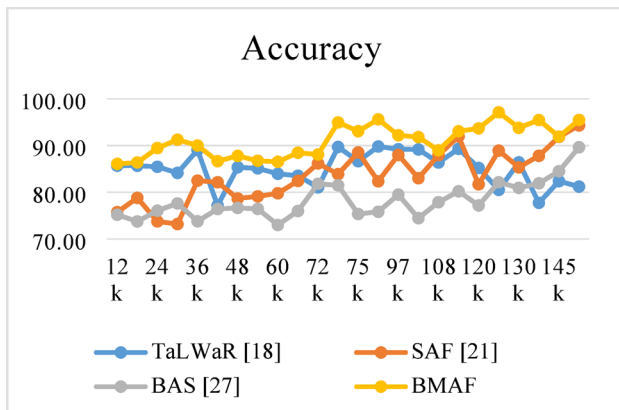


Fig. 7 Accuracy for authentication requests

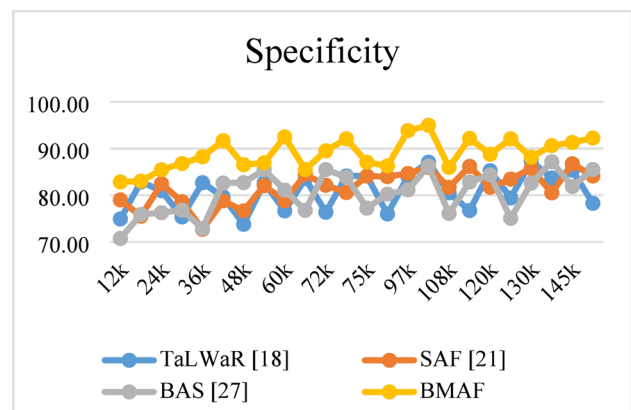


Fig. 9 Specificity for authentication requests

5 Results and discussions

The objective of evaluating the proposed framework, MABF-IoT, was to assess its effectiveness and QoS performance under attack scenarios. The primary performance metrics were evaluated using the models from [18, 21], and [27]. The proposed framework compares QoS performance with models [30, 34], and [38]. A fair comparison is ensured by testing each model under identical conditions. The results are shown in Figs. 4 to 11, which analyze the performance of each model.

5.1 Model evaluation based on primary metrics

The model is subjected to several test sample requests, NTS (Number of Test Samples), varying from 12k to 150k, to simulate different operational loads.

Figure 6 indicates that MABF-IoT consistently maintains a higher precision rate, generally above 95%, whereas SAF& BAS fluctuate around 90% TO 95%, and TaLWaR dips below 85%. Figure 7. illustrates that MABF-IoT consistently maintains a stable accuracy, generally above 90%,

whereas SAF fluctuates around 85% to 95%, BAS slightly lowers to 80% to 90%, and TaLWaR dips below 80%.

Figure 8 indicates that MABF-IoT consistently maintains a stable recall level, generally above 90%, whereas SAF fluctuates between 85% and 90%, BAS slightly decreases to 80% to 85%, and TaLWaR dips below 80%. Figure 9 shows the Specificity levels of the compared models. MABF-IoT consistently maintains a stable specificity level, generally above 90%, whereas SAF fluctuates between 80% and 90%, BAS slightly decreases to 75% to 85%, and TaLWaR dips below 80%.

The empirical data show that MABF-IoT enhances precision and accuracy compared to existing models. Its high and consistent precision and accuracy make it very reliable for correctly identifying authorized devices. Likewise, MABF-IoT's strong and consistent recall rate and specificity ensure it is a dependable method for accurately detecting unauthorized devices. The primary assessment of MABF-IoT's performance indicates that it has the potential to replace the current authentication system, offering a scalable, effective, and secure solution for IoT authentication.

5.2 Model evaluation based on QoS metrics

In this section, the model is evaluated based on its QoS metrics. The model undergoes attack request communications NC (number of attack requests in communications), ranging from 5k to 52k, to represent different levels of cyberattacks.

The higher the identification delay, the slower the system responds, which affects IoT operations. Figure 10 shows that MABF-IoT consistently has the lowest identification delay, generally around 140ms, while BPPS varies between 150ms and 160ms. BSDCE has a slightly higher delay of 160ms to 170ms, and PuF exceeds 180ms. A lower authentication delay indicates a more efficient and responsive system. Figure 11 demonstrates that MABF-IoT maintains the lowest authentication delay, staying below 3ms. In comparison, BPPS fluctuates between 4ms and 6ms, BSDCE ranges from 5ms to 7ms, and PuF has the highest delay, spanning from 6ms to 9ms.

High energy consumption quickly depletes power resources, leading to system failures or disruptions. Figure 12 shows that MABF-IoT has the lowest energy use, staying below 3 mJ. In comparison, BPPS varies between 4 mJ and 6 mJ, BSDCE has a slightly higher range of 5 mJ to 7 mJ, and PuF consumes the most energy, ranging from 7 mJ to 11 mJ. Figure 13 demonstrates that MABF-IoT achieves the highest throughput among the models, consistently ranging from 1000 kbps to 1500 kbps. Meanwhile, PuF fluctuates between 1000 kbps and 1200 kbps, BPPS has a slightly lower throughput of 800 kbps to 1200 kbps, and BSDCE has the lowest, ranging from 700 kbps to 1100 kbps. Figure 14 illustrates the packet delivery ratio (PDR) of the compared models, with MABF-IoT maintaining the highest PDR, consistently above 85% to 95%. Conversely, BPPS varies between 80% and 90%, BSDCE has a slightly lower range of 75% to 85%, and PuF records the lowest PDR, below 85%.

The empirical data show that MABF-IoT outperforms existing models. MABF-IoT's low and stable identification and authentication delays, along with lower energy consumption, indicate faster response times in device identification and authentication, and enhance the battery life efficiency of IoT devices. The higher throughput and PDR demonstrate that MABF-IoT is more efficient, sustainable, and reliable.

The comparative analysis indicates that MABF-IoT outperforms all the other models across primary performance metrics and QoS parameters. MABF-IoT provides a comprehensive and advanced solution for fast and accurate device identification and authentication, thereby enhancing the security, efficiency, and reliability of the IoT environment.

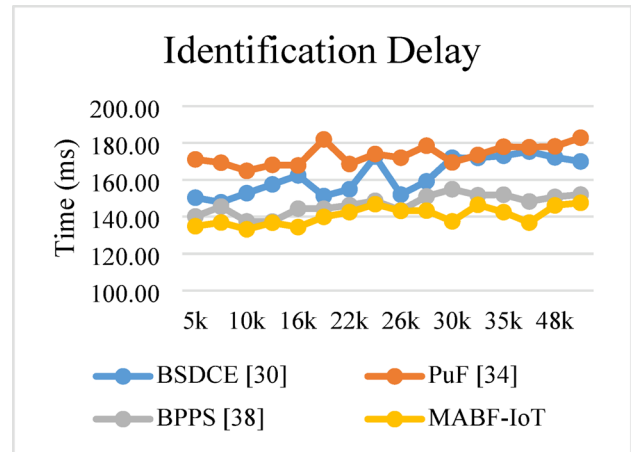


Fig. 10 Identification delay

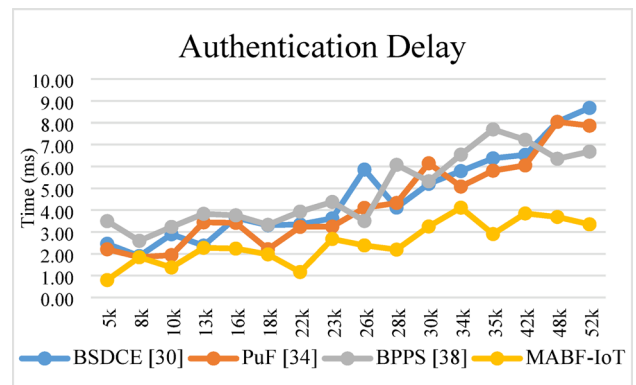


Fig. 11 Authentication delay

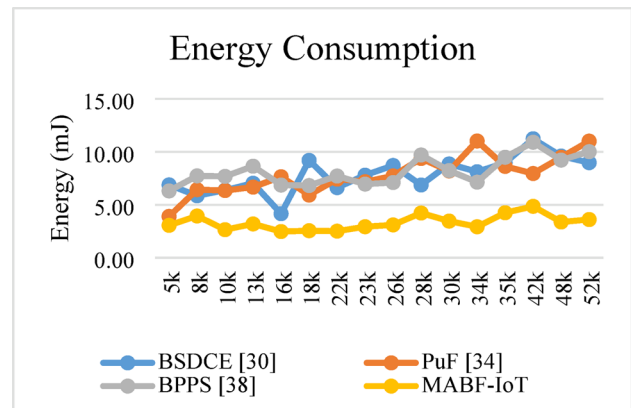


Fig. 12 Energy consumption

6 Conclusion and future work

The proliferation of IoT devices in modern networks has greatly increased security concerns, especially regarding mutual authentication and anomaly detection within M2M communication. Protecting critical infrastructure from malicious identity and credential threats demands robust

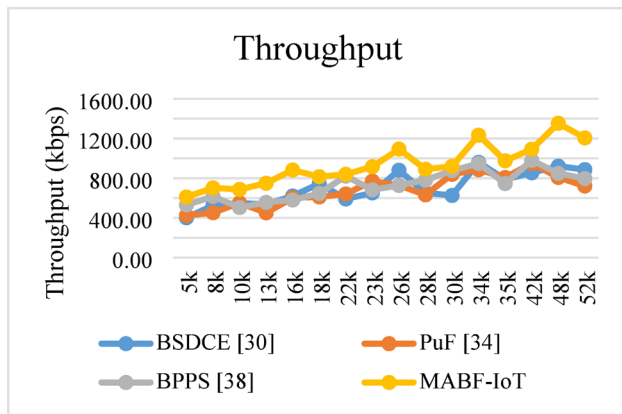


Fig. 13 Throughput

identification and authentication methods. The proposed MABF-IoT framework addresses these threats by integrating decentralized authentication using blockchain, an intelligent anomaly detection system based on GNN, and digital fingerprinting of devices. This comprehensive system ensures effective detection of abnormal behavior and provides tamper-proof, unique identifiers for secure device authentication. MABF-IoT was thoroughly tested with four IoT security datasets: BoT-IoT, IoT-23, Risk-Based Authentication, and IoT Intrusion Detection. Benchmarking against various existing authentication and anomaly detection models demonstrated that the MABF-IoT framework surpasses these models in several areas, including higher authentication accuracy, enhanced threat detection, and more efficient resource use in constrained IoT environments. The empirical results support the advantages of MABF-IoT in both identification and authentication processes, achieving faster times while utilizing minimal computational resources.

In follow-up studies, the framework will be extended to various devices across more domains and operating environments.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval Not applicable.

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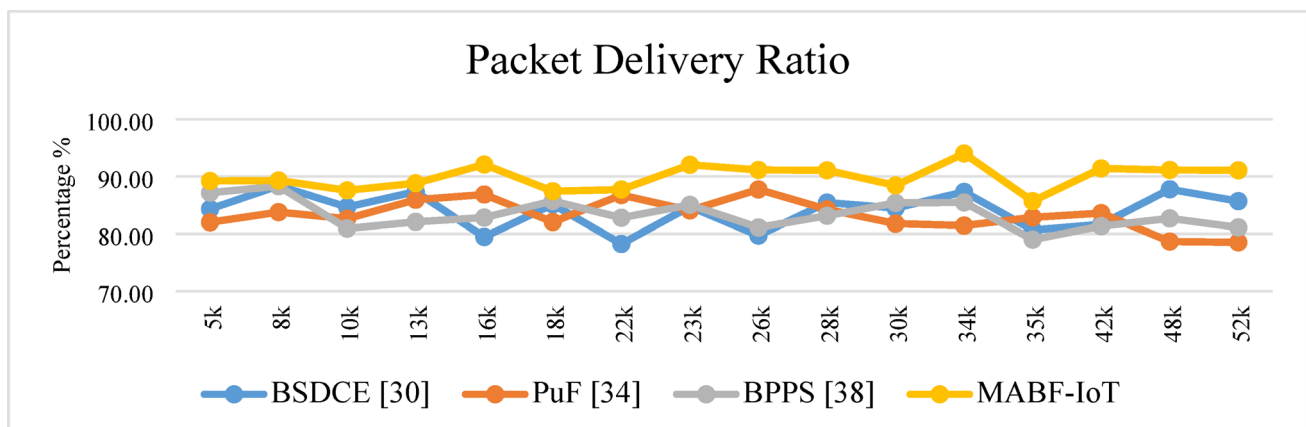


Fig. 14 Packet delivery ratio

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